

1) Reinforcement learning is a type of ML in which an agent learns by interacting with an environment and receives rewards or penalties for its actions.

Main objective:- To learn an optimal policy that maximizes total cumulative reward over time.

2) A Markov decision process is a mathematical framework used to model decision making where outcomes depend on current state and action.

components:-

- S - Set of states
- A - Set of actions
- P - State transition probability
- R - Reward
- γ - Discount factor.

3) The Bellman equation for state-value function is:

$$V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma V(s')]$$

It expresses value of state as expected reward plus.

4) Exploration

Tries new actions

Gains more knowledge

may give lower short-term reward.

Exploitation

chooses best known action

maximizes immediate reward.

Gives higher short-term reward.

5) Multi Armed Bandit problem is a decision making problem where an agent chooses among multiple actions

to maximize total reward while balancing exploration & exploitation.

- 6) DP is a method for solving optimization problems by breaking them into smaller subproblems.

Applications

- * policy evaluation
- * policy improvement
- * policy iteration.

- 7) TD learning is a RL method that updates value estimate using current reward & estimated value of next state.

TD(0) update eqn:-

$$V(S_t) = V(S_t) + \alpha [R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$

- 8) SARSA

on-policy algorithm
more conservative
uses next action

Q-learning
off-policy algorithm.
learns optimal policy faster.
uses max Q-value.

- 9) A DQN uses a deep neural network to approximate Q-value function.

Advantages:-

- * Handles large states spaces.
- * Generalizes to unseen states.

- 10) DDQN is an improved version of DQN uses separate networks for action selection & action evaluation.

Why preferred over DQN

- * Reduces Q-value overestimation
- * Improves learning stability
- * Gives better performance & accuracy.